

Introduction

This lab introduces students to the fundamentals of network design using **Cisco Packet Tracer**, a powerful network simulation tool. The goal of this experiment is to understand how to create, configure, and test computer networks using both **switches** and **routers**. The lab is divided into two major parts:

1. Building a **local network** using **switches**.
2. Building an **interconnected network** using **routers**.

Objectives

- To learn the basic interface and tools of Cisco Packet Tracer.
- To understand how to connect multiple computers using a switch.
- To configure IP addresses and test communication using ping.
- To learn how to interconnect different networks using routers.
- To verify successful communication between networks.

Equipment and Software

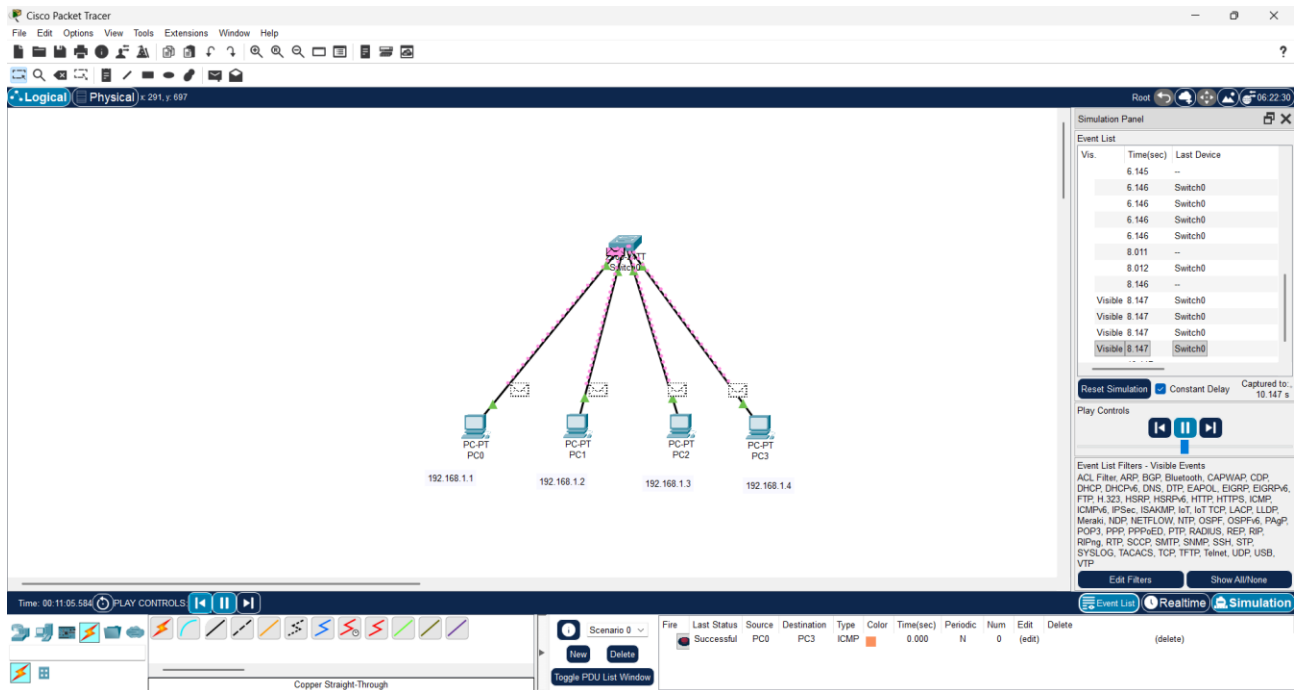
- **Software:** Cisco Packet Tracer (latest version)
- **Devices Used:**
 - 4 PCs (PC0, PC1, PC2, PC3)
 - 2 Switches (Switch0, Switch1)
 - 2 Routers (Router0, Router1)
 - Copper Straight-Through and Cross-Over Cables

Part 1: Building a Local Network Using Switch

Procedure:

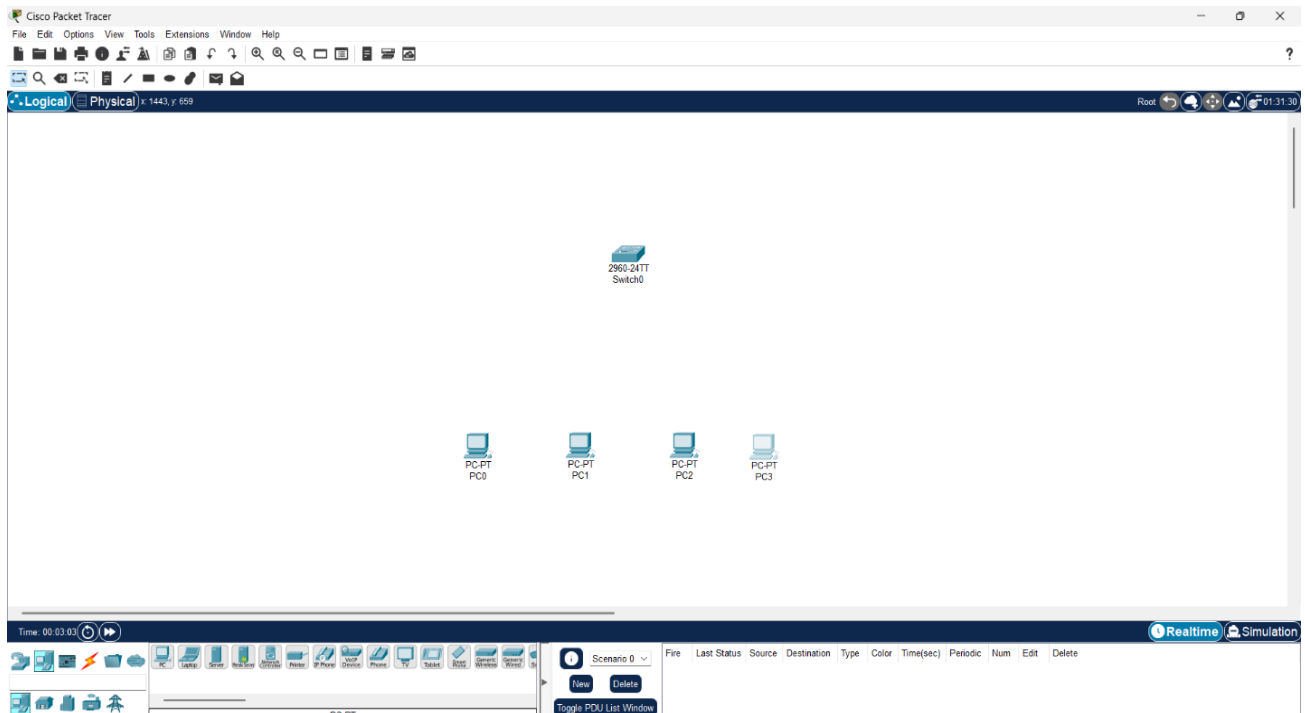
- **Open Cisco Packet Tracer:**

Launch the application and open a new workspace.



- **Add Devices**

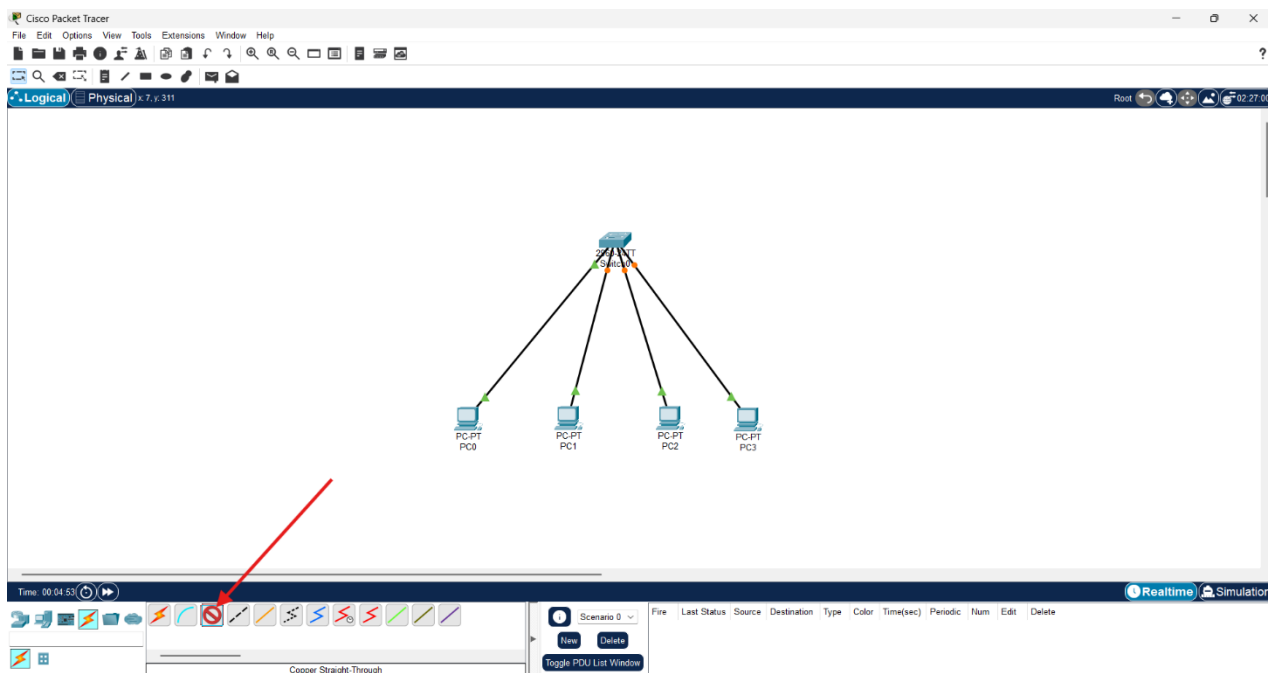
Drag and drop one switch (Switch0) and four PCs onto the workspace



- **Connect Devices**

Use **Copper Straight-Through Cables** to connect each PC to the switch.

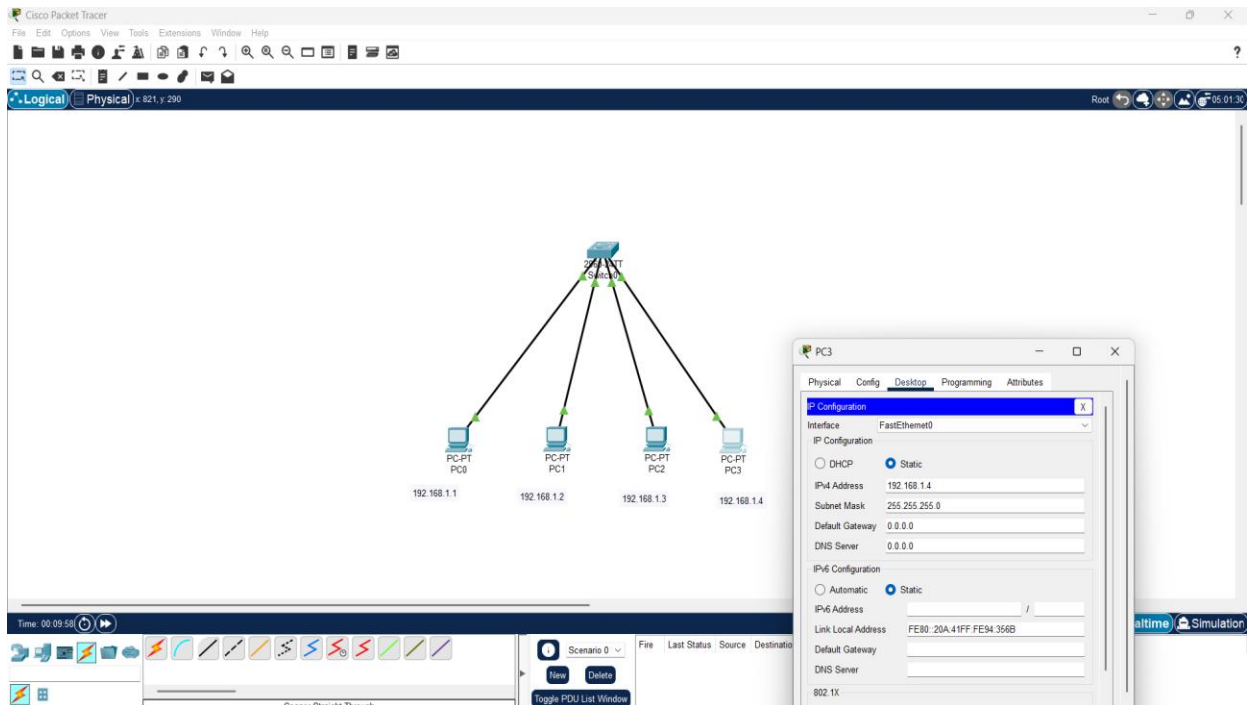
- PC0 → Switch0 (FastEthernet0/1)
- PC1 → Switch0 (FastEthernet0/2)
- PC2 → Switch0 (FastEthernet0/3)
- PC3 → Switch0 (FastEthernet0/4)



- **Assign IP Addresses**

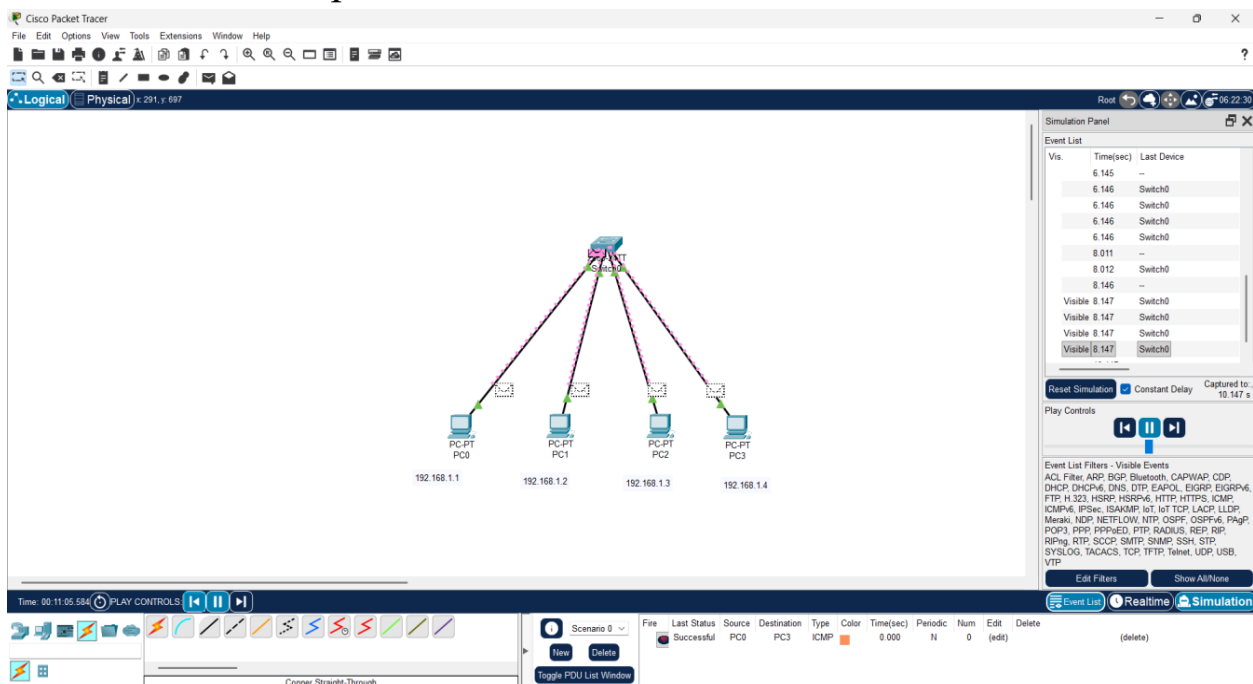
Go to each PC → Desktop → IP Configuration, and assign the following:

- PC0: 192.168.1.1 / 255.255.255.0
- PC1: 192.168.1.2 / 255.255.255.0
- PC2: 192.168.1.3 / 255.255.255.0
- PC3: 192.168.1.4 / 255.255.255.0



- **Test Connectivity**

Test The Connectivity By sliding the ICMP packets in each PC and then running the simulation. Here is the figure of the Successful Network setup.



- **Result**

All PCs were able to communicate successfully with each other through the switch. The switch efficiently created a local area network (LAN).

- **Conclusion**

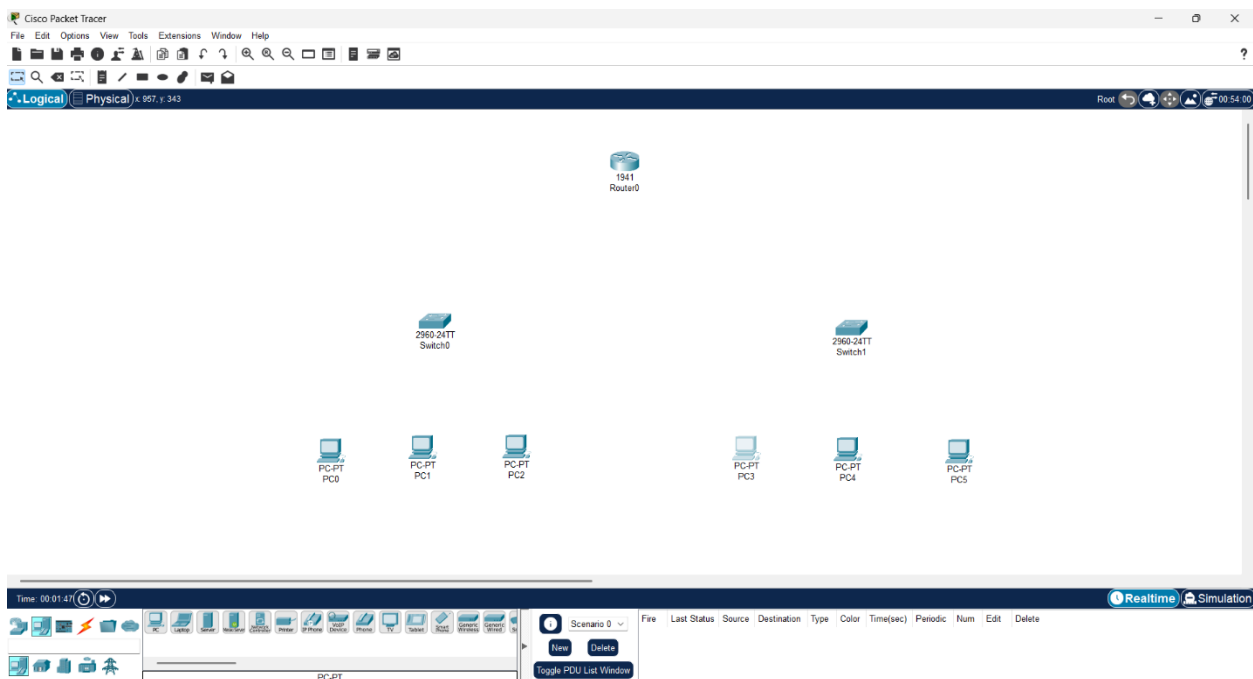
This part demonstrated how to design and configure a simple LAN using a switch. The experiment verified the role of a switch in connecting multiple devices within the same network.

Part 2: Building a Network Using Routers

Procedure:

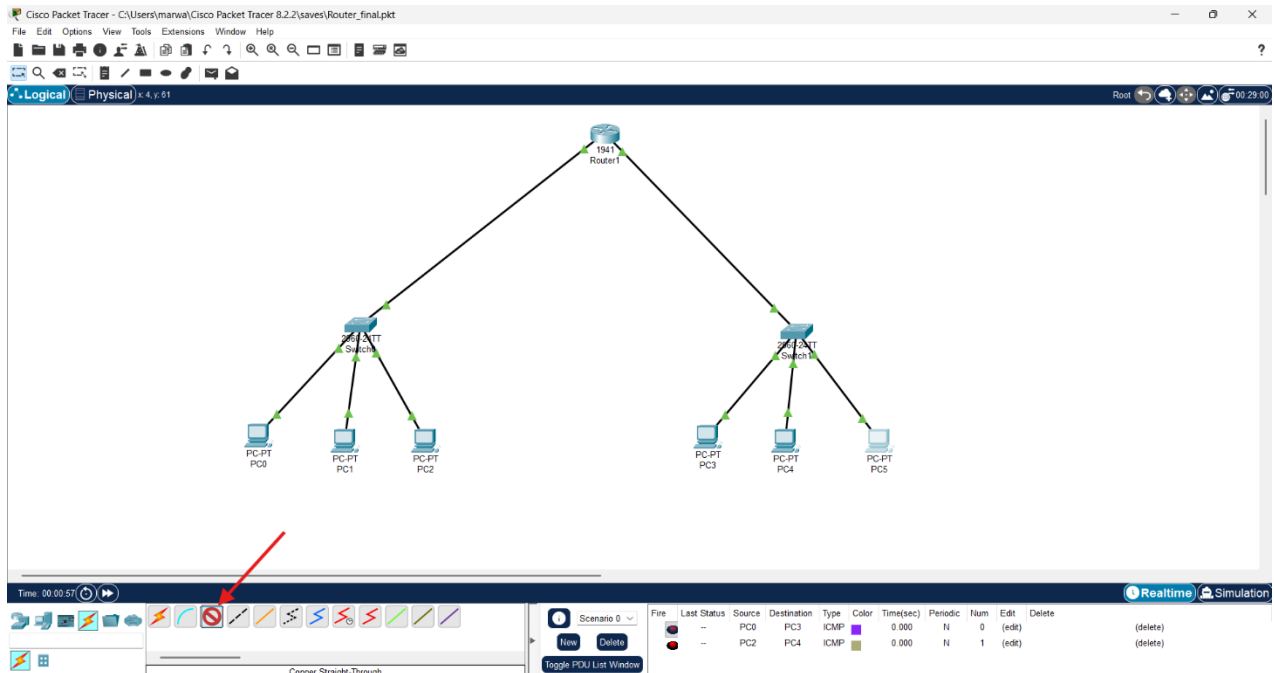
- **Add Devices**

Drag and drop **a router, two switches, and six PCs** onto the workspace.



- **Connect Devices**

- PC0, PC1 & PC2 → Switch0
- PC3, PC4 & PC5 → Switch1
- Switch0, Switch1 → Router1 (FastEthernet0/0, FastEthernet0/1)



- **Add Devices**

- **Network 1:**

- PC0: 10.10.10.1 / 255.255.255.0
- PC1: 10.10.10.2 / 255.255.255.0
- PC2: 10.10.10.3 / 255.255.255.0

- **Network 2:**

- PC3: 192.168.0.1 / 255.255.255.0
- PC4: 192.168.0.2 / 255.255.255.0
- PC5: 192.168.0.3 / 255.255.255.0

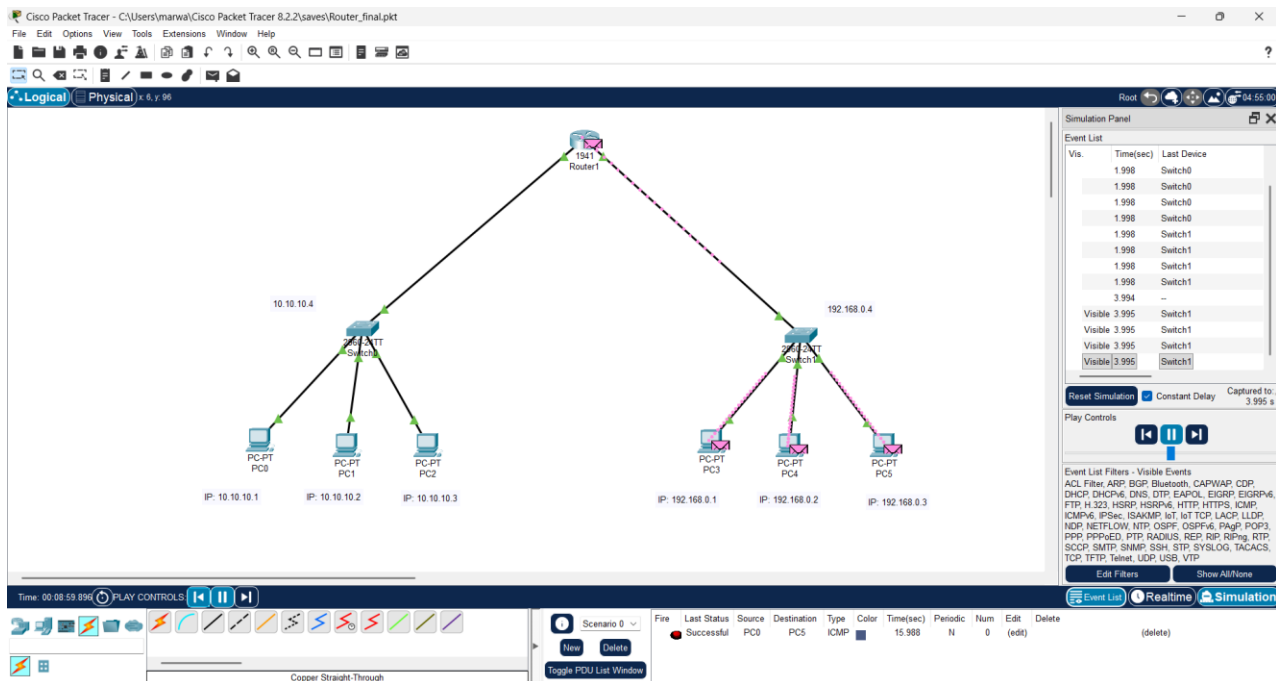
- **Switch Interconnection:**

- Switch0: 10.10.10.4
- Switch1: 192.168.0.4

Add These Two IP addresses in the Router also in the Gateway Section of the corresponding Connected PC's.

- **Test Connectivity**

Test The Connectivity By sliding the ICMP packets from a pc connected with the first switch to a pc that is connected with the second Switch (for example: From the PC0→PC5 and then running the simulation. Here is the figure of the Successful Network setup.



Result

All PCs from both networks communicated successfully through the routers. Static routing enabled data exchange between the two networks.

Conclusion

This part demonstrated how routers connect multiple networks and route packets between them. The experiment confirmed that routers are essential for **inter-network communication**, while switches are used for **intra-network communication**.

Final Discussion

Through this lab, students gained practical experience in designing and configuring basic computer networks using Cisco Packet Tracer. The experiment helped differentiate the roles of switches and routers:

- **Switch:** Used within a single LAN for device interconnection.
- **Router:** Used to connect multiple LANs and manage network traffic between them.